



DRAWING API · BACKDROP

Use the Paint Editor to draw a brown dirt backdrop with 6 dark oval holes



STEPS TO FOLLOW

1. Open Scratch → delete cat sprite
2. Click backdrop thumbnail → "Paint" (pencil icon)
3. Fill canvas brown
4. Ellipse tool dark brown → draw 6 ovals in 3×2 grid
5. Click back-arrow to return

TIP

Keep your hole ovals consistent in size
— you'll position mole clones to
match each hole centre in later steps.

DID YOU KNOW?

You just used a 2D drawing API — the same API that browsers use for HTML Canvas, and that Figma uses for its design canvas.



OBJECT TEMPLATES

Add a hedgehog or mole sprite, size it to 50, and hide it on flag click



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE: when  clicked —

hide

TIP

If you can still see the mole, make sure the Code tab is on the MOLE sprite (blue outline in sprite panel).

DID YOU KNOW?

A hidden sprite with "when I start as a clone" scripts is exactly like a class — it defines behaviour without doing anything visible until instantiated.



RANDOM POSITIONING · CLONES

Make the mole teleport to a random hole position and create one clone



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE master: when clicked —

hide

go to x: (pick random (-160) to (160)) y: (-60)

create clone of [myself]

TIP

Adjust the x range to match your painted holes. If holes are at x=-120, 0, 120 use pick random (-120) to (120).

DID YOU KNOW?

"pick random" is a pseudo-random number generator — it produces numbers that feel random but are actually deterministic sequences computed from a seed.



TIMER • OBJECT LIFECYCLE

The mole clone appears, waits 1.5 seconds, hides, waits 0.5 seconds, then deletes itself



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE: when I start as a clone —

show

wait (1.5) seconds

hide

wait (0.5) seconds

delete this clone

TIP

Try wait (0.8) for a faster pop-up or
wait (2.5) for a slower, easier game.

This timer controls difficulty!

DID YOU KNOW?

This show → wait → hide → delete
pattern is the game-object lifecycle:
create → active → deactivate →
destroy. Used in every game engine
for temporary objects.



MOUSE EVENTS · INPUT

Add a click handler on the Mole that plays a sound when
whacked



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE: *when this sprite clicked* →

play sound [pop] until done

TIP

Click the Sounds tab on the Mole sprite to browse and add sound effects. "Chomp", "Zap", or "Pop" all work well.

DID YOU KNOW?

"when this sprite clicked" is `addEventListener("click")` in JavaScript — the event pattern that powers every button, menu, and interactive element on the entire web.



VARIABLES · REWARD STATE

Create a Score variable and increment it on each successful click



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE: *when this sprite clicked* →

play sound [pop] until done

change [Score] by (1)

delete this clone

TIP

Right-click the Score display and choose "Large readout" for a satisfying big counter.

DID YOU KNOW?

Every leaderboard, XP system, in-game currency, and achievement tracker is built on this same pattern:

event detected → variable updated → display refreshed.



PARALLEL EXECUTION • CONCURRENCY

Make the master mole spawn new clones continuously in a forever loop



SCRATCH BLOCKS — ON THE MOLE SPRITE

— MOLE master: when  clicked —

hide

set [Score] to (0)

forever

wait (1.2) seconds

go to x: (pick random (-160) to (160)) y: (-60)

create clone of [myself]

TIP

Lower the wait time (0.8 secs) for more moles at once. Higher (2 secs) gives beginners more breathing room.

DID YOU KNOW?

Two "when  clicked" scripts running at the same time is concurrency — the same concept behind `async/await` in JavaScript and Python's `threading` module.



COUNTDOWN TIMER

Add a 30-second countdown Timer variable that ticks down once per second



SCRATCH BLOCKS — TIMER SCRIPT

when  clicked (timer script)

set [Timer] to (30)

repeat (30)

wait (1) seconds

change [Timer] by (-1)

TIP

Increase to 45 or 60 seconds for an easier game. The "repeat" count always matches the start Timer value.

DID YOU KNOW?

This repeat-wait-change pattern is how every countdown works in code — from quiz timers to rocket launch sequences.



STATE MACHINES • PROGRAM TERMINATION

When the timer hits zero, show a message and stop the game



SCRATCH BLOCKS — AFTER THE REPEAT LOOP

(after the repeat loop)

```
say [Time's up! 🔔] for (2) secs
```

```
stop [all]
```

💡 TIP

Share your game on Scratch — click

"Share" at the top. Friends can try to beat your high score!

🧠 DID YOU KNOW?

PLAYING → GAME_OVER is a state machine. Login flows, vending machines, traffic lights, and payment systems all work as state machines.